PROGRAM 13 – MINIMAX ALGORITHM FOR GAMING

AIM

To write a Python program to implement the Minimax algorithm for gaming to find the best possible move.

ALGORITHM

1. Start the program.
2. Create a game tree with terminal values.
3. Consider two players: MAX and MIN.
4. MAX tries to obtain the maximum score.
5. MIN tries to obtain the minimum score.
6. Evaluate the terminal nodes.
7. Apply the Minimax algorithm recursively.
8. Return the best value to the parent node.
9. Display the best move and its value.

CODE

```
def minimax(depth,node_index,max_player,values,max_depth):
    if depth==max_depth:
        return values[node_index]
    if max_player:
        return
max(minimax(depth+1,node_index*2,False,values,max_depth),minimax(depth+1,node_index*2+1,F
alse,values,max_depth))
    else:
        return
min(minimax(depth+1,node_index*2,True,values,max_depth),minimax(depth+1,node_index*2+1,Tru
e,values,max_depth))
values=[3,5,2,9,12,5,23,23]
depth=3
result=minimax(0,0,True,values,depth)
print("Terminal Values:",values)
print("Best Value using Minimax:",result)
```

INPUT

```
values=[3,5,2,9,12,5,23,23]
depth=3
```

OUTPUT

```
Terminal Values: [3, 5, 2, 9, 12, 5, 23, 23]
Best Value using Minimax: 12
```

RESULT

Thus, the Minimax algorithm was successfully implemented in Python to determine the best possible move in a game tree.